

Xiang Li

+1 765 637 5476 ▪ li2068@purdue.edu ▪ [Google Scholar](#) ▪ [Linkedin Profile](#)

CAREER SUMMARY

Ph.D. student specializing in the end-to-end optimization of Large Language Model applications. Expertise in designing high-performance system, from novel agent memory architectures to low-latency LLM inference pipeline. Focused on bridging the gap between complex model architectures and efficient, scalable production deployments.

EDUCATION

Purdue University, West Lafayette 2021–2027 (expected)

- Ph.D. in Computer Engineering. GPA. 3.74
 - Advisor: Saurabh Bagchi
 - Research Area: System for ML, Efficient LLM Serving, Agent Memory

Purdue university, West Lafayette 2015–2019

- B.S. Computer Engineering. GPA. 3.89

WORK EXPERIENCE

ML Algorithm Intern- Futurewei Technology Inc. San Jose, CA Aug 2025 - Dec 2025

- Developed a novel memory architecture utilizing Graph Neural Networks to address the limited context window and degraded attention result in long-horizon embodied task planning.
- Designed a dynamic experience graph mechanism to consolidate past agentic experiences with current observations, facilitating more robust state representation and decision-making.

ML Research Intern- Houston Methodist Research Institute. Houston, TX Jun 2023 – Aug 2023

- Built a 3D group equivariant CNN-based segmentation model for non-contrast CT brain scans, reaching 87% soft tissue segmentation accuracy and reducing diagnostic latency for ischemic stroke detection.
- Partnered with physicians to translate AI models into clinical pipelines, accelerating imaging-based diagnosis.

Control Software Engineer- Cummins Inc. Columbus, IN Jun 2019 – Jul 2021

- Engineered dual-encryption bootloader security for Cummins ECM, eliminating unauthorized firmware tampering across 250k+ vehicles in production.
- Collaborated with cross-functional teams to deliver model-based control software for critical automotive components, ensuring reliability in high-volume manufacturing.
- Tuned and validated 30+ OBD parameters for RAM truck compliance with U.S. emission standards, strengthening regulatory approval and system robustness.

Graduate Teaching Assistant-Purdue University. West Lafayette, IN Aug 2021 – July 2025

- Teaching assistant for Purdue ECE embedded software senior design course ECE 47700.
- Teaching assistant for Purdue ECE embedded system design course ECE 36200.
- Teaching assistant for Purdue First Year Engineering course ENG 131,132.

Graduate Research Assistant-Purdue University. West Lafayette, IN Summer 2022, Summer 2024

- Engineered a unified scheduling system for heterogeneous edge devices and hybrid networks, optimizing resource allocation across diverse computing nodes.
- Established a testbed integrating Private 5G and Wi-Fi 6 to evaluate network performance and reliability under industrial constraints.

TECHNICAL SKILLS

Languages. C, Python, R, JS ▪ **Machine Learning.** PyTorch, TensorFlow, LLM Inference Optimization
▪ **H/W Platforms.** Jetson AGX, Jetson NX, Jetson Nano, Arduino Uno, Raspberry Pi 4 (baremetal) ▪ **Web Technologies.** socketio, Wireshark, React

COURSEWORK

Graduate level. Estimation Theory, Random Process, Computational Modeling, Deep Learning, Reinforcement Learning Theory

PUBLICATIONS	[ICML'26]	Embodied Task Planning via Graph-Informed Action Generation with Large Language Model Xiang Li, Ning Yan, Masood Mortazavi (2026)	
	[NIPS@DL4C'26]	Deep-Reproducer: From Paper Understanding to Code Generation Pengcheng Chen, Ning Yan, Zihan Zhao, Yixiao Lin, Huaibo Chen, Yue Hu, Qinbo Bai, Xiang Li, Masood Mortazavi (2025)	
	[arXiv]	Ascendra: Dynamic Request Prioritization for Efficient LLM Serving. Xiang Li*, Azam Ikram*, Sameh Elnikety, Saurabh Bagchi. arXiv preprint (2025)	
	[ESOC'25]	Enhanced Brain Tissue Segmentation In Non-Contrast CT For Ischemic Stroke Diagnosis Xiang Li, Saurabh Bagchi, John Volpi, Stephen Wong, Kelvin Wong (2025)	
	[arXiv]	HopTrack: A Real-time Multi-Object Tracking System for Embedded Devices Xiang Li, Cheng Chen, Yuan-yao Lou, Mustafa Abdallah, Kwang Taik Kim, Saurabh Bagchi. arXiv preprint (2024)	
	[arXiv]	Dynamic DAG-application scheduling for multi-tier edge computing in heterogeneous networks Xiang Li, Mustafa Abdallah, Yuan-Yao Lou, Mung Chiang, Kwang Taik Kim, Saurabh Bagchi. arXiv preprint (2023)	
	[SRDS'22]	DAG-based task orchestration for edge computing Xiang Li, Mustafa Abdallah, Shikhar Suryavansh, Mung Chiang, Kwang Taik Kim, Saurabh Bagchi (2022)	
ACHIEVEMENTS	Tinker Research Grant , Thinking Machine Lab.		2025
	▪ Awarded \$5000 credits for research project uses Tinker.		
	Honor Society Eta Kappa Nu , Purdue University.		2021
	▪ Selected for being in the top 10% of graduate students.		
	Eli Shay Scholarship , Purdue University		2017
SERVICE	Reviewer , ICML 26'		2026
	Reviewer , KDD 26'		2026
	Reviewer , BHI 25'		2025
	Student Administrator , USENIX ATC 24'		2024
	Artifact review committee , OSDI 24'		2024
	Artifact review committee , USENIX ATC 24'		2024
	Reviewer , IEEE Networking Letter		2023